

JavaScript Roadmap: Basic to Advanced

1. Introduction to JavaScript

- What is JavaScript?
- History and Versions
- How JavaScript works
- Setting up Environment (Browser, Editor)
- Embedding JavaScript in HTML

2. JavaScript Basics

- Statements and Syntax
- Variables (var, let, const)
- Data Types (Primitive and Reference)
- Operators (Arithmetic, Comparison, Logical, etc.)
- Type Conversion and Coercion

3. Control Structures

- Conditional Statements (if, else, switch)
- Loops (for, while, do-while)
- Break and Continue

4. Functions

- Function Declaration and Expression
- Arrow Functions
- Parameters and Arguments
- Return Statement
- Callback Functions

5. Objects and Arrays

- Object Literals and Properties

- Accessing and Modifying Objects
- Nested Objects
- Arrays and Array Methods
- Multidimensional Arrays

6. DOM Manipulation

- What is DOM?
- Selecting Elements (getElementById, querySelector, etc.)
- Modifying Elements (textContent, innerHTML, etc.)
- Adding/Removing Elements
- Events and Event Listeners

7. Advanced JavaScript

- Closures
- Scope and Lexical Environment
- Hoisting
- The 'this' Keyword
- Prototypes and Inheritance
- ES6+ Features (Spread, Rest, Destructuring)

8. Asynchronous JavaScript

- setTimeout and setInterval
- Callback Hell
- Promises
- Async/Await
- Fetching API using fetch()

9. Error Handling and Debugging

- try-catch-finally
- Throwing Errors

- Console Methods
- Debugging in Browser

10. JavaScript in Browser

- Browser Object Model (BOM)
- Window Object
- Timers
- Dialogs (alert, confirm, prompt)
- Storage (localStorage, sessionStorage)

11. Modules and Tooling

- ES6 Modules (import/export)
- Webpack Basics
- Babel
- NPM and Package Management

12. Testing

- Introduction to Testing
- Unit Testing with Jest
- Testing DOM Manipulations

13. JavaScript and APIs

- REST APIs
- JSON
- Using APIs with fetch() and Axios

14. Object-Oriented JavaScript

- Constructor Functions
- Classes and Objects
- Inheritance using Classes
- Encapsulation, Polymorphism

15. Functional Programming Concepts

- Pure Functions
- Higher-Order Functions
- Immutability
- Map, Filter, Reduce

16. Real-World Projects and Practice

- ToDo List App
- Weather App using API
- Calculator
- Form Validation Project